



Assessment of Heavy metal pollution and its health implications in groundwater for drinking purpose around inactive mines, SW region of Cuddapah Basin, South India

Y. Sudharshan Reddy^{a,*}, V. Sunitha^b

^a Department of Civil Engineering, Sai Rajeswari Institute of Technology, Proddatur, Kadapa 516362, India

^b Department of Geology, Yogi Vemana University, Kadapa 516005, India

ARTICLE INFO

Keywords:

Groundwater
Ecological risk assessment
Heavy metal pollution index
Human health risk
Southwestern part of Cuddapah Basin

ABSTRACT

Trace metal leachate comes from inactive mine reservoirs in the study area. The main intention of the study was to analyze the metal content in groundwater samples around inactive mines in the southwest of the Cuddapah Basin and to assess the risk to the local habitants based on calculations using several statistical methods. In this context, the study emphasizes the use of an integrated heavy metal pollution approach, ecological risk assessment incorporating potential health risks, and a multivariate statistical approach at an inactive mining site in the southwestern Cuddapah Basin in southern India. In 2019, we collected 100 samples from bore wells in both pre & post monsoon seasons and ten heavy metals (As, Co, Cd, Fe, Cr, Ni, Mn, Pb, Zn, and Sr) were analyzed using an Agilent 725 ICP-OES instrument. The results showed that As (0–60.5 and 0–56.3 µg/L), Cd (0–28.1 and 0–31.1 µg/L), Ni (0–110.2 and 0–99 µg/L), Pb (0–175.6 and 0–92.3), Sr concentration (0–1150 and 0–1440 µg/L) exceeded permissible values in both seasons. Heavy metal pollution index (HPI), heavy metal evolution index HEI, and pollution degree DOC are used to evaluate the metal pollution of drinking water in this area. As per the HPI values, 17.5 % and 10% of samples show a high pollution index; as per the classes of HEI, 95 % and 100% of samples show a low pollution class during both seasons. According to DOC, all groundwater samples fall in the low contamination zone. As per ecological risk classification, 90% of the samples cause extreme to high ecological risk in both seasons in the study area. From the perspective of health risk assessment, both adults and children have non-carcinogenic effects in the study area; and adequate remedial procedures or treatments are required to avoid metal pollution of groundwater.m.

Introduction

Heavy metals are a group of metallic elements that have a high density and are toxic to humans and other living organisms in certain concentrations. Some of the most common heavy metals include lead, mercury, cadmium, arsenic, and chromium. Heavy metals can enter the environment through a variety of sources, including industrial and agricultural activities, transportation, and waste disposal. They can contaminate soil, water, and air, and can accumulate in the food chain. Humans can be exposed to heavy metals through various pathways, including inhalation of polluted air, ingestion of contaminated food and water, and dermal contact with contaminated soil or water. Once inside the body, heavy metals can cause a variety of health problems, including neurological damage, developmental delays, kidney damage, and

cancer. Preventing exposure to heavy metals is important for protecting public health and the environment. This can be achieved through a variety of strategies, including regulations on industrial and agricultural practices, proper waste disposal, and monitoring of air, water, and soil quality (Sudharshan Reddy et al., 2020a; Sudharshan Reddy et al., 2020b). Heavy metals are dispersed into the biological field through anthropogenic activities; like as industrial activity, mining, agriculture, and transportation activities. In this case, waste materials containing metal-rich sulfides from mining activities have been stored or discarded. Annoying activities such as urbanization, industrialization, transportation, misuse of fertilizers, pesticides, improper disposal of sewage and toxic chemical solid waste, and natural processes like precipitation, erosion, and weathering of crustal materials increase the heavy metals in soil and water (Abraham and Susan, 2017; Prathap et al., 2019). Certain

* Corresponding author.

E-mail address: yenugusudharshan@gmail.com (Y. Sudharshan Reddy).

<https://doi.org/10.1016/j.totert.2023.100069>

Received 22 April 2023; Received in revised form 4 July 2023; Accepted 28 July 2023

Available online 29 July 2023

2772-8099/© 2023 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).